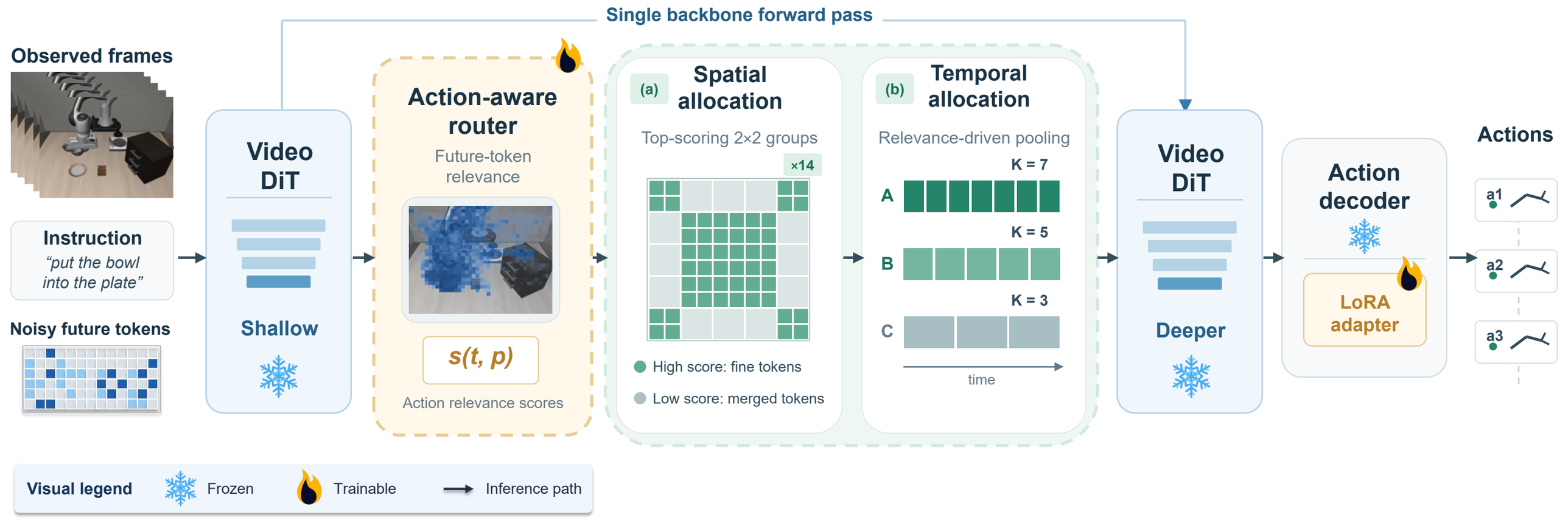

Single backbone forward pass
Observed frames
Instruction
“put the bowl into the plate”
Noisy future tokens
Video DiT
Shallow
Action-aware router
Future-token relevance
s(t, p)
Action relevance scores
(a)
Spatial allocation
Top-scoring 2×2 groups
×14
High score: fine tokens
Low score: merged tokens
(b)
Temporal allocation
Relevance-driven pooling
K = 7
A
K = 5
B
K = 3
C
time
Video DiT
Deeper
Action decoder
LoRA adapter
Actions
a1
a2
a3
Visual legend
Frozen
Trainable
Inference path